

## APUTITEEQ, Greenland

WMO: 043510

Lat: 67.783N

Lon: 32.285W

Elev: 23

StdP: 101.05

Time Zone: -3.00 (GRL)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 3             | -20.9      | -18.6 | -26.0                         | 0.4 | -19.3 | -23.9 | 0.4 | -17.0 | 21.1                  | -3.0 | 17.9 | -6.7 | 2.5                   | 50   | 0.933 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |     |      |     |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|-----|------|-----|------|---------------------|------|-----|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%  |      | 2%  |      | 0.4%                |      | 1%  |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB  | MCWB | DB  | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e) | (f)  | (g) | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                  | (p)  |
| 8             | 4.0                    | 8.5             | 5.1  | 7.3 | 4.5  | 6.4 | 4.0  | 5.8                 | 7.4  | 5.0 | 6.5  | 4.4 | 5.9  | 3.3                  | 330  |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 4.7                             | 5.3 | 5.8  | 4.0 | 5.0 | 5.1  | 3.3 | 4.8 | 4.6  | 20.3          | 7.5  | 18.7 | 6.6  | 17.6 | 6.0  | 10.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 16.6          | 12.9 | 10.4 | DB                         | -23.0 | 11.4               | 3.6 | 1.3                                                | -25.6 | 12.4       | -27.7 | 13.1       | -29.7 | 13.9       | -32.2 | 14.8 | (4) |
| (5)               |      |      | WB                         | -23.6 | 7.6                | 3.5 | 1.4                                                | -26.1 | 8.6        | -28.1 | 9.4        | -30.1 | 10.2       | -32.6 | 11.2 | (5) |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov  | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)  | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | -2.8   | -7.4  | -8.9  | -9.4  | -5.0  | -1.0  | 0.8   | 3.0   | 3.4   | 2.2   | -1.0  | -3.7 | -6.8 |
|                                                                               | DBStd       | 5.67   | 5.06  | 5.29  | 4.72  | 3.92  | 2.37  | 2.15  | 2.30  | 1.82  | 1.98  | 2.52  | 3.01 | 4.21 |
|                                                                               | HDD10.0     | 4664   | 539   | 528   | 602   | 450   | 342   | 276   | 216   | 205   | 233   | 342   | 410  | 521  |
|                                                                               | HDD18.3     | 7706   | 797   | 762   | 860   | 700   | 601   | 526   | 474   | 463   | 483   | 601   | 660  | 779  |
|                                                                               | CDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0    |
|                                                                               | CDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0    |
|                                                                               | CDH23.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0    |
|                                                                               | CDH26.7     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0    |
| Wind                                                                          | WSAvg       | 3.5    | 4.5   | 4.3   | 4.0   | 3.6   | 3.0   | 2.4   | 2.4   | 2.5   | 3.2   | 4.1   | 4.2  | 4.1  |
| Precipitation                                                                 | PrecAvg     | 921    | 99    | 97    | 89    | 69    | 50    | 53    | 36    | 73    | 91    | 82    | 93   | 90   |
|                                                                               | PrecMax     | 1306   | 220   | 288   | 184   | 147   | 125   | 125   | 84    | 110   | 240   | 133   | 252  | 255  |
|                                                                               | PrecMin     | 596    | 32    | 22    | 18    | 18    | 15    | 24    | 7     | 28    | 26    | 38    | 16   | 10   |
|                                                                               | PrecStd     | 197    | 52    | 60    | 46    | 34    | 23    | 21    | 22    | 22    | 46    | 26    | 53   | 59   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 3.9   | 2.1   | 1.2   | 5.6   | 6.2   | 7.6   | 11.1  | 9.2   | 8.6   | 6.4   | 4.2  | 3.1  |
|                                                                               |             | MCWB   | -0.1  | -0.2  | -1.6  | 2.9   | 2.4   | 4.4   | 7.0   | 4.9   | 4.7   | 2.8   | 0.9  | -0.4 |
|                                                                               | 2%          | DB     | 1.2   | 0.7   | -0.2  | 2.9   | 4.6   | 6.2   | 8.9   | 7.9   | 6.7   | 4.4   | 2.0  | 1.4  |
|                                                                               |             | MCWB   | -0.8  | -0.7  | -1.8  | 1.2   | 1.8   | 3.6   | 5.9   | 4.8   | 3.5   | 2.5   | 0.4  | -1.0 |
|                                                                               | 5%          | DB     | 0.1   | -0.2  | -1.3  | 1.5   | 3.2   | 5.1   | 7.6   | 7.0   | 5.5   | 3.4   | 0.9  | -0.1 |
|                                                                               |             | MCWB   | -1.2  | -1.1  | -2.6  | 0.3   | 1.2   | 3.0   | 5.0   | 4.4   | 3.3   | 1.9   | -0.3 | -1.4 |
|                                                                               | 10%         | DB     | -1.0  | -1.6  | -2.7  | 0.6   | 2.2   | 4.1   | 6.5   | 6.1   | 4.7   | 2.4   | 0.0  | -1.2 |
|                                                                               |             | MCWB   | -2.1  | -2.7  | -3.8  | -0.3  | 0.6   | 2.4   | 4.3   | 4.0   | 2.9   | 0.9   | -0.9 | -2.4 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 0.8   | 0.6   | -0.3  | 3.2   | 3.2   | 5.1   | 8.3   | 6.5   | 5.7   | 4.0   | 1.7  | 0.9  |
|                                                                               |             | MCDB   | 1.7   | 1.3   | 0.3   | 4.6   | 4.8   | 7.0   | 10.0  | 7.8   | 6.5   | 5.0   | 3.1  | 2.2  |
|                                                                               | 2%          | WB     | 0.0   | 0.0   | -1.3  | 1.5   | 2.2   | 4.0   | 6.3   | 5.5   | 4.7   | 3.1   | 0.7  | -0.2 |
|                                                                               |             | MCDB   | 1.0   | 0.4   | -0.6  | 2.5   | 3.8   | 5.6   | 8.3   | 6.8   | 5.6   | 4.0   | 1.5  | 0.5  |
|                                                                               | 5%          | WB     | -0.9  | -1.2  | -2.5  | 0.6   | 1.5   | 3.2   | 5.3   | 4.8   | 4.1   | 2.2   | 0.0  | -1.3 |
|                                                                               |             | MCDB   | -0.2  | -0.3  | -1.5  | 1.3   | 2.8   | 4.7   | 7.1   | 6.2   | 5.0   | 3.1   | 0.6  | -0.3 |
|                                                                               | 10%         | WB     | -2.1  | -2.6  | -3.8  | -0.1  | 0.9   | 2.5   | 4.5   | 4.3   | 3.4   | 1.2   | -0.7 | -2.3 |
|                                                                               |             | MCDB   | -1.1  | -1.7  | -2.9  | 0.5   | 1.9   | 3.9   | 6.2   | 5.7   | 4.3   | 2.1   | 0.0  | -1.2 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 5.1   | 5.6   | 6.1   | 5.5   | 4.5   | 4.4   | 4.6   | 4.0   | 3.4   | 2.8   | 3.0  | 4.3  |
|                                                                               |             | MCDBR  | 5.2   | 5.2   | 6.2   | 5.8   | 5.9   | 5.7   | 6.3   | 5.2   | 4.7   | 4.0   | 4.3  | 4.9  |
|                                                                               |             | MCWBR  | 4.1   | 4.1   | 5.2   | 4.2   | 3.8   | 3.7   | 3.8   | 3.1   | 2.8   | 2.8   | 3.1  | 4.1  |
|                                                                               | 5% WB       | MCDBR  | 5.2   | 5.0   | 5.9   | 5.0   | 5.1   | 5.1   | 5.7   | 4.6   | 3.9   | 3.5   | 3.8  | 4.6  |
|                                                                               |             | MCWBR  | 4.9   | 4.3   | 5.2   | 3.7   | 3.6   | 3.5   | 3.9   | 3.1   | 2.8   | 2.6   | 2.9  | 4.2  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        |        | 0.202 | 0.229 | 0.267 | 0.313 | 0.320 | 0.320 | 0.322 | 0.328 | 0.304 | 0.256 | N/A  | N/A  |
|                                                                               | taud        |        | 1.837 | 2.049 | 2.143 | 2.039 | 2.073 | 2.202 | 2.352 | 2.376 | 2.341 | 2.120 | N/A  | N/A  |
|                                                                               | Ebn at Noon |        | 90    | 582   | 774   | 824   | 863   | 877   | 866   | 816   | 736   | 523   | N/A  | 0    |
|                                                                               | Edh at Noon |        | 11    | 49    | 82    | 122   | 132   | 120   | 100   | 88    | 69    | 45    | N/A  | 0    |
| All-Sky Solar<br>Radiation                                                    | RadAvg      |        | 0.04  | 0.54  | 2.17  | 4.55  | 6.84  | 7.49  | 6.38  | 4.40  | 2.26  | 0.81  | 0.10 | 0.00 |
|                                                                               | RadStd      |        | 0.00  | 0.07  | 0.14  | 0.27  | 0.33  | 0.45  | 0.36  | 0.22  | 0.26  | 0.08  | 0.01 | 0.00 |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page